BT404 Midterm Current Solved 2022

Research Methods and Skill Enhancement (Virtual University of Pakistan)



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BT404 Food Biotechnology

Midterm 2022

Current Papers Solved

We Grow Together ❤

Define plasticity of Fats?

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not flow at room temperature and are thus solid fats. The spreading quality of butter is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of one kind of a molecule.

What is the equation of Ash?

$$\%\ \text{ash (dry basis)} = \frac{\text{wt after ashing} - \text{tare wt of crucible}}{\text{original sample wt} \times \text{dry matter coefficient}} \times 100$$

Sequence of Events in Food Borne Diseases ?

A sequence of events happens in all food borne diseases. An understanding of these sequences is helpful in investigating the cause (the source and means of transmission) of a food borne disease. It also helps in recognizing how the sequence can be broken.

- Initially, there has to be a source of a pathogen.
- The pathogen has to contaminate a food.
- Consumption of the food contaminated with a pathogenic virus or parasites.
- It lead to viral or parasitic infection.

Define foodborne disease?

Foodborne disease is caused by consuming contaminated foods or beverages. Many different disease-causing microbes or pathogens can contaminate foods, so there are many different types of foodborne diseases. Most foodborne diseases are infections caused by a variety of bacteria, viruses, and parasites.

What is normal range of protein and fats in meat?

The average protein content of meat ranges from 16 to 23 % and the average fat content ranges from 10 to 40%.

Name any two optional ingredients used for cheese making?

- Sodium benzoate
- Sodium propionate

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- Sorbic acid
- Hydrogen peroxide

What is the Ideal Solvent For Fat Extraction?

- High solvent power for lipids
- Low solvent power for nonlipids
- No residue
- Evaporate easily
- Low boiling point
- Non flammable / not explosive
- Nontoxic
- Low surface tension with food
- Single compound
- Non-hygroscopic

What are the Ingredients used for Cheese Making?

- Milk
- Calcium Chloride
- Starter Culture
- Adjunct Culture
- Colour
- Enzyme
- Salt
- Optional ingredients

Write two types of wheat ?

There are two types of wheat:

- Hard wheat (High Protein)
- Soft wheat (Low Protein)

Write the used products of wheat ?

Common used products of wheat

- Vermicelli
- Broken wheat (Dalia)
- Noodles
- Suji
- Cereals
- Canned soups
- Pasta

Which factors affecting Microbial growth?

Factor affecting Microbial growth are

- pH

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- Moisture
- Humidity of environment
- Temperature
- Availability of Oxygen

What are the Heat control methods to control Microbial growth?

Heat control methods to control Microbial growth :

High Temperature :

- Pasteurization
- Sterilization
- UHT etc

Low Temperature :

- Blast freezing
- Cryogenic freezing etc.

Name of bacteria in yogurt?

Lactic Acid Bacteria (LAB)

Why are fats used in food ?

The body uses fat as a fuel source, and fat is the major storage form of energy in the body. Fat also has many other important functions in the body, and a moderate amount is needed in the diet for good health.

Write a note on Listeria monocytogens?

Listeria monocytogens: are Gram-positive, motile bacteria. They do not form spores, but are quite resistant to desiccation and heat. Found in the soil and associated with birds and animals. Cause disease called listeriosis, which can result in septicaemia, meningitis encephalitis and spontaneous abortions in pregnant females.

Write the types of botulism?

There are five types of botulism:

- Food-borne botulism
- Wound botulism
- Infant botulism
- Adult infectious botulism
- Inadvertent botulism

Cereals are used for many purposes what it’s types?

Most common cereal foods include:

- Rice
- Wheat
- Maize (corn)

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- Millets such as:
  • Jowar
  • Bajra
  • Ragi

Can be processed further to make:

- Flour
- Starch
- Oil
- Bran
- Syrup
- Sugar

Write the types of corn grown?

Many types of corn are grown:

- Sweet corn
- Popcorn
- Dent corn
- Flour corn

What are the types of rice ?

Rice is available as

- Milled
- Unpolished
- Polished
- Parboiled
- Faked
- Puffed

Write down the steps of Refining of crude oils?

Refining is a 5 step process

- Degumming
- Neutralising
- Washing and Drying
- Bleaching
- Deodourising

How infra red dying process is done?

Infra red lamps with a filament temperature of 2000-2500 K can penetrate heat to food and shorten drying time to 20-25 minutes only.

What is phenylethylamine?

- Found in chocolate

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- Vasopressor amine
- Constrict blood vessels
- Can cause migraine in susceptible person.

What are the types of lipids?

They are classified as

- Simple lipids
- Complex lipids
- Derived lipids

Explain Agar Plating ?

Plating surface spread-plate technique rather than with pour plates for maximal exposure of the cells to atmospheric oxygen and to avoids heat stress from molten agar. Agar spread plates should be dried overnight before being inoculated.

Mention at least three names of Microbes that pasteurization inhibits the growth?

- Bacillus
- Micrococci
- Lactococci
- Arthrobacter
- Microbacteria
- Corynebacteria

What are the limitations of UHT processing?

The main limitations of UHT processing are the

- Cost
- Complexity of the plant which arises from the necessity to
- Sterilise packaging materials
- Pipework/tanks
- Maintenance of sterile air and surfaces in filling machines.
- Higher skill levels required by operators and staff

What are the functions of food substances when ingesting by living organisms?

Foods substances that are ingested by living organisms and can be metabolized to perform following functions:

- Source of energy
- Source of raw materials for growth and development
- To supply minute amount of nutrients to regulate vital metabolic processes
- To supply food components (phytochemicals) that retard the development and progression of degenerative diseases.

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Foods can be fermented in three different ways based on the sources of the desirable microorganisms. What are they?

- Natural fermentation
- Back slopping
- Controlled fermentation

Define Dietary fiber?

Dietary fiber or roughage is the portion of plant-derived food that cannot be completely broken down by human digestive enzymes. Dietary fiber includes the nondigestible carbohydrates. These may be either water soluble or water-insoluble. The water insoluble group, which includes wheat products and wheat bran. The water-soluble fibers, includes such as some cereal brans and pectin.

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